

Performance Analysis of Weighted A* Search Applied to the 4×4 Sliding Puzzle (15-Puzzle)

Course	CSE 318 – Artificial Intelligence Sessional
Assignment	Offline 1: Solving the N-Puzzle using A* Search
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Heuristic Used	Manhattan Distance

1. Introduction

A* solves the N-Puzzle optimally by expanding states in order of $f(n) = g(n) + h(n)$. Weighted A* changes this to $f(n) = g(n) + W \cdot h(n)$, $W \geq 1.0$, trading optimality for speed as W grows. This report measures that trade-off on ten solvable 4×4 puzzles using the Manhattan distance heuristic, for $W \in \{1.0, 1.2, 2.0, 5.0\}$.

Manhattan Distance Heuristic

For each tile, Manhattan distance sums the vertical and horizontal moves needed to reach its goal position, ignoring other tiles; the heuristic is the total over all tiles. It is admissible (never overestimates true cost) since it only counts unavoidable moves, and it is more informed than Hamming distance, giving A* stronger pruning.

2. Methodology

Weighted A* was run on each of the ten 4×4 puzzles for $W = 1.0, 1.2, 2.0, 5.0$ with the Manhattan heuristic, recording solution cost (moves in the returned path) and nodes expanded (states dequeued and processed). $W = 1.0$ is standard, optimal A*, used as the baseline.

3. Metrics Table

Cost and nodes expanded for each puzzle across all four weight values.

Puzzle	W = 1.0 Cost / Nodes	W = 1.2 Cost / Nodes	W = 2.0 Cost / Nodes	W = 5.0 Cost / Nodes
1	46 / 921,585	46 / 121,871	48 / 1,028	88 / 1,786
2	48 / 653,289	50 / 123,968	56 / 4,197	86 / 1,154
3	50 / 1,531,550	50 / 110,227	58 / 6,682	86 / 5,347
4	43 / 1,398,278	43 / 202,760	45 / 22,143	75 / 2,296
5	40 / 92,142	40 / 7,199	50 / 24,445	78 / 11,972
6	44 / 767,081	44 / 77,587	54 / 23,292	78 / 13,166
7	53 / 1,213,394	57 / 431,741	71 / 18,295	119 / 12,978
8	52 / 322,339	52 / 54,785	74 / 20,602	118 / 21,549

Puzzle	W = 1.0 Cost / Nodes	W = 1.2 Cost / Nodes	W = 2.0 Cost / Nodes	W = 5.0 Cost / Nodes
9	45 / 348,088	45 / 18,702	55 / 4,701	81 / 490
10	41 / 578,891	41 / 56,080	41 / 455	71 / 5,696
Average	46.2 / 782,664	46.8 / 120,492	55.2 / 12,584	88.0 / 7,643

Table 1: Solution cost and nodes expanded (Manhattan heuristic) for ten 4×4 puzzles across $W = 1.0, 1.2, 2.0, 5.0$.

4. Graphical Analysis

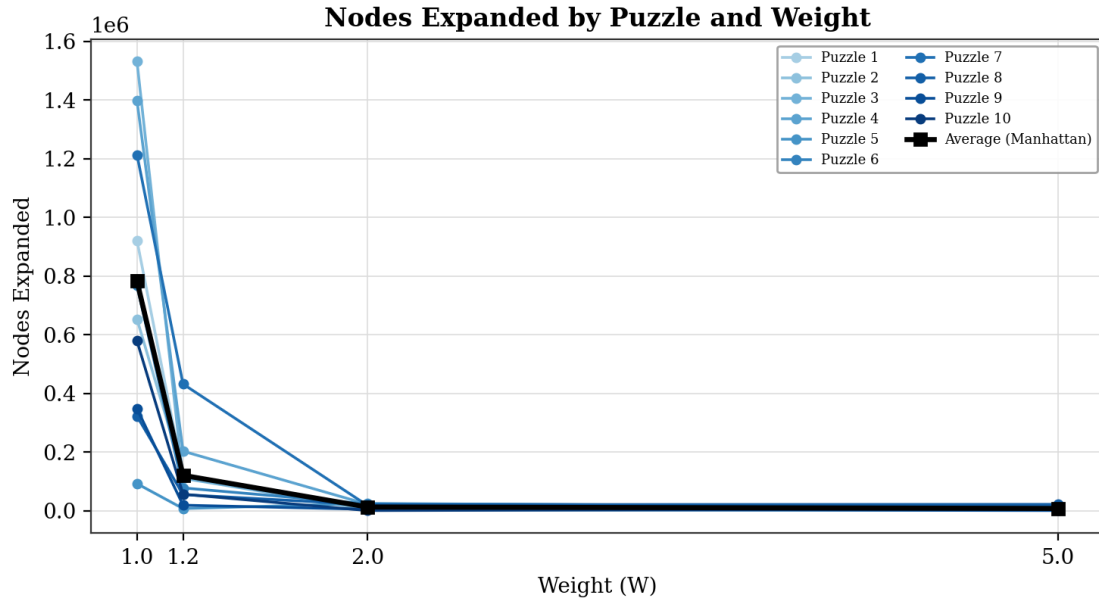


Figure 1: Nodes expanded vs. weight, per puzzle.

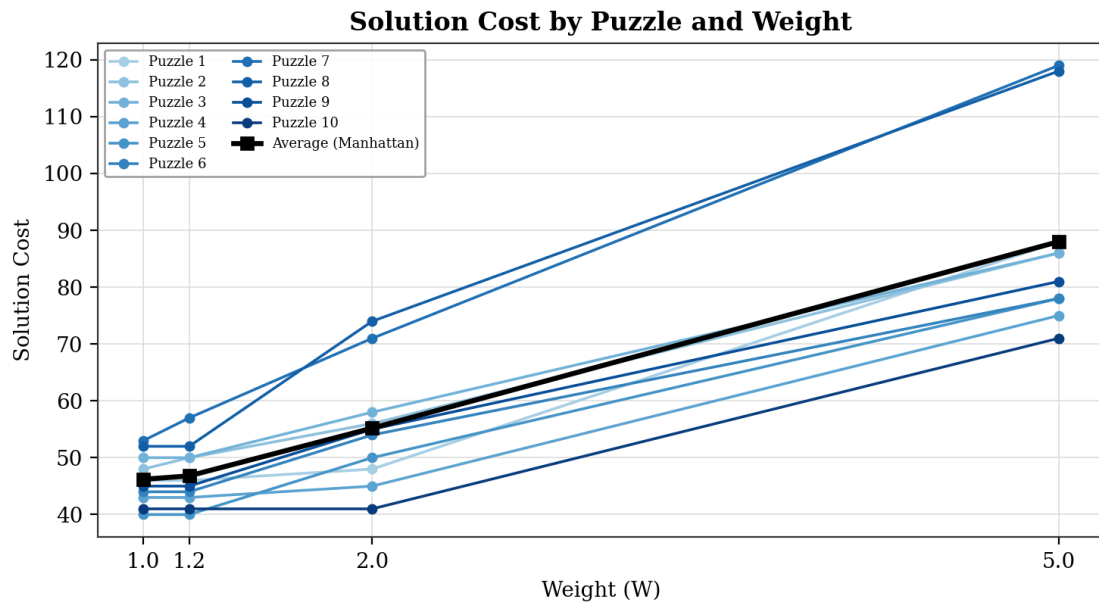


Figure 2: Solution cost vs. weight, per puzzle.

5. Discussion

Nodes expanded. $W = 1.0$ explores by far the most nodes (avg. $\sim 782,700$), with Puzzle 3 exceeding 1.5 million. $W = 1.2$ already cuts this to $\sim 120,500$ (over 80% reduction); $W = 2.0$ falls to $\sim 12,600$, and $W = 5.0$ to $\sim 7,600$. Even small weight increases prune the search sharply, since the heuristic term dominates the priority function.

Solution cost. Average cost rises from 46.2 ($W=1.0$) to 46.8 ($W=1.2$), 55.2 ($W=2.0$), and 88.0 ($W=5.0$). Most puzzles show no cost increase at all from $W=1.0$ to $W=1.2$ (e.g. Puzzles 1–6, 9, 10), so this step is nearly free. Beyond $W=2.0$, cost degrades sharply, nearly doubling by $W=5.0$.

Trade-off. Small weights (1.0–1.2) keep cost near-optimal while cutting nodes substantially, making $W=1.2$ an attractive default. Larger weights (2.0, 5.0) trade further efficiency for noticeably worse solutions. Puzzles 7 and 8, the hardest under plain A^* , also show the steepest cost growth as W increases.

6. Conclusion

Weighted A^* trades optimality for speed: nodes expanded drop sharply as W increases, most steeply between $W=1.0$ and $W=1.2$, while cost stays near-optimal only for small W and worsens noticeably for $W \geq 2.0$. $W=1.2$ offers the best balance; $W=5.0$ is fastest but least accurate.